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Accurate $PM_{2.5}$ urban air pollution forecasting using multivariate ensemble learning Accounting for evolving target distributions

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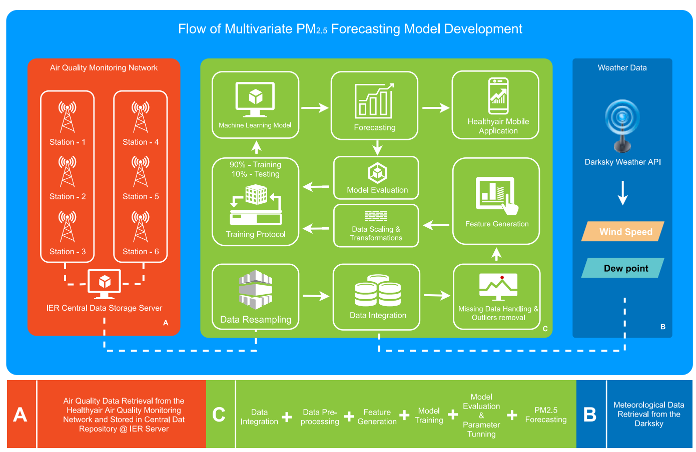
HIGHLIGHTS

- The proposed model accurately predicts urban $PM_{2.5}$ contamination for the next 24 h.
- MI-based correlation analysis is valuable for understanding the influence of meteorological variables on $PM_{2.5}$ concentration.
- Combining rolling mean features of $PM_{2.5}$ with meteorological features improves $PM_{2.5}$ prediction accuracy significantly.
- The forecasting model's training protocol effectively addressed the data's non-stationarity.

GRAPHICAL ABSTRACT



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ABSTRACT

Over the past decades, air pollution has caused severe environmental and public health problems. According to the World Health Organization (WHO), fine particulate matter ($PM_{2.5}$), a key component reflecting air quality, is the fourth leading cause of death worldwide after cardiovascular disease, smoking, and diet. Various research efforts have aimed to develop $PM_{2.5}$ forecasting models that can be integrated into a solution to mitigate the adverse effects of air pollution. However, $PM_{2.5}$ forecasting is challenging because air pollution data are non-stationary and influenced by multiple random effects. This paper proposes an effective multivariate multi-step ensemble machine learning model for predicting continuous 24-h $PM_{2.5}$ concentrations, considering meteorological conditions, the rolling mean of $PM_{2.5}$ time series, and temporal features. $PM_{2.5}$ is strongly correlated with space and time. Therefore, forecasting results from one location are insufficient to represent the level of air pollution for an entire city. In this study, we established six real-time air quality monitoring sites in different regions, including traffic, residential, and industrial areas in Ho Chi Minh City (HCMC), and generated

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forecasting results for each station. Various statistical methods are incorporated to evaluate the performance of the model. The experimental results confirm that the model performs well, substantially improving its forecasting accuracy compared to existing $PM_{2.5}$ forecasting models developed for HCMC. In addition, we analyze to determine the contribution of different feature groups to model performance. The model can serve as a reference for citizens scheduling local travel and for healthcare providers to provide early warnings.

1. Introduction

Air pollution is affecting people's health, the environment, and the economy (Pandey et al., 2021). Transportation, industrial operations, coal power plants, and the use of solid fuels are all key factors in Vietnam's growing air pollution. According to the WHO, 2.2 million people die each year in western Pacific countries, with Vietnam alone accounting for sixty thousand fatalities due to air pollution (WHO, 2018). As of 2020 (World Bank, 2021), the urban population in Vietnam was around 37% of the total population. The urbanization process, increased traffic, and inefficient industrial setup all contribute to increased air pollution in the cities.

$PM_{2.5}$ is an important component of the Air Quality Index (AQI) (Zhao et al., 2018). When $PM_{2.5}$ concentrations are above the WHO's threshold, it is hazardous to the environment and people's health (WHO, 2021). Scientific and epidemiological research has linked increased short-term $PM_{2.5}$ exposure to an increase in hospital admissions, mostly for respiratory and cardiovascular problems (Xing et al., 2016; Dominici et al., 2006). Long-term exposure to $PM_{2.5}$ may also be related to considerably higher death rates, mostly owing to heart disease and lung cancer, according to Franklin et al., (2008).

According to the study by Vu et al. (2020), there are 1135 premature deaths in HCMC each year owing to illnesses associated with $PM_{2.5}$ air pollution - lung cancer, cardio-pulmonary, and Ischemic Heart Diseases. Furthermore, Nhung et al. (2022) investigated the mortality burden owing to long-term $PM_{2.5}$ exposure in Hanoi, Vietnam, and found an extra 2696 recorded fatalities per year attributed to air pollution (95% CI: 2225 to 3158).

As a result, reliable $PM_{2.5}$ forecasting is required to assist residents in planning their daily schedules to avoid the adverse effects of air pollution on their health. Several conventional methodologies, including statistical and deterministic methods, have been widely employed for ambient air quality modelling throughout the last decade (Liao et al., 2021). These methods aim to determine the likelihood of different outcomes instead of predicting the exact outcome.

Accurate air quality prediction is a challenging task mainly due to the volatility characteristics of $PM_{2.5,}$ including non-stationarity, randomness, and uncertainty in the data. The air quality also differs from one place to another place, e.g. from traffic areas to residential and industrial areas across the city. Machine learning methods have recently been widely adopted in air quality prediction research due to their higher prediction performance, accelerated data processing speed, and support for auto-updates with new data. Rakholia et al., 2022 examined a variety of machine learning approaches for daily and hourly $PM_{2.5}$ forecasting. Appendix A (Table 1) summarizes machine learning-based case studies for $PM_{2.5}$ prediction. The forecasting results vary because the level of air pollution varies by geolocation, which is also affected by meteorological conditions.

Furthermore, Xiao et al. (2020) created a WLSTME (Weighted Long Short-term Memory neural network Extended model) model to forecast the average daily $PM_{2.5}$ concentration at a given station taking into account the disparate placement of monitoring sites. WLSTME has shown excellent power to simulate the impact of wind and geographical distance on such dependencies, which is a need for addressing both spatial and temporal dependencies concurrently. Zhong et al. (2021) provided an overview of four critical applications of ML use in environmental research and engineering, which include prediction, feature importance extraction, anomaly detection, and material/chemical discovery. They also mentioned that ML algorithms might be excellent tools for automatically categorizing and analyzing chemical patterns in environmental contaminants that would be difficult to identify manually. One significant benefit of ML algorithms is their ability to detect trends or patterns in data without human interaction.

However, the influence of different input features and covariate feature selection based on non-linear correlations with $PM_{2.5}$ prediction has not yet been examined. The goal of this research is to create a powerful machine learning model for multi-step hourly air quality $PM_{2.5}$ forecasting for HCMC utilizing a powerful tree-based ensemble machine learning method that incorporates meteorological variables, rolling mean of $PM_{2.5}$ time series, and temporal features.

1.1. *Climate of Ho Chi Minh city*

Ho Chi Minh City experiences a rainy season from May to November and a dry season from December to April. The city is influenced by two main wind patterns: the West-Southwest monsoon from June to October with an average speed of 3.6 m/s, and the North-Northeast monsoon from November to February at 2.4 m/s (Le et al., 2022; Khoi et al., 2021).

Predominant wind regimes shift seasonally, with southwest monsoon winds prevailing during the rainy season, bringing heavy rainfall and higher humidity, while northeast monsoon winds dominate the dry season, resulting in drier and cooler conditions. The average air temperature is 27.9 °C (from 1980 to 2019). The average lowest temperature (27.3 °C) is usually witnessed in January and the highest in April about 30.4 °C (Le et al., 2022). The annual rainfall was about 1951 mm in the period 1980 to 2019 (Le et al., 2022), with 90% occurring during the rainy season from May to November, and the highest rainfall typically in June and September (Khoi et al., 2021).

This climatic pattern significantly influences $PM_{2.5}$ levels, as the heavy rains during the wet season can help reduce airborne particulates by washing them out of the atmosphere, whereas the dry season can see higher concentrations of $PM_{2.5}$ due to reduced precipitation and increased air stagnation.

2. Data and methods

2.1. *Study area*

The study area covers the urban area of HCMC Appendix B (Fig. 1), the primary economic center and biggest city in Vietnam, with a population density of 4292/km^2. The city contributes more than 22% of Vietnam's GDP (Nguyen, 2019). However, the serious air pollution problem cannot be ignored to protect the urban environment and human health. Air pollution in the city is increasing mainly due to increasing traffic and industrial setup in urban areas (Danh, 2020). Therefore, this study adopted HCMC as the area for developing an air quality $PM_{2.5}$ forecasting model and deployment on mobile applications for the benefit of citizens. The study period for this research is from March 2021 to September 2022.

2.2. *Data*

$PM_{2.5}$ air pollution data was acquired from the Air Quality Monitoring Network (AQMN) established by the Healthyair project team across HCMC in February 2021. The AQMN has six monitoring stations situated around the city, covering traffic, residential, and industrial zones. The data was collected at a 1-min sample rate. In addition, (Rakholia et al., 2023) detail the steps involved in gathering data as well as the technical specifications of air quality monitoring equipment and the process flow for getting air quality data. The location of each air pollution monitoring station, together with its corresponding number, is shown in Appendix B (Fig. 2). The weather conditions significantly impact on $PM_{2.5}$ air quality (Chen et al., 2020). Meteorological variables including temperature, humidity, dewpoint, wind speed, ultraviolet radiation, and visibility were all collected from Darksky weather service (Darksky API). The key benefit of using the Darksky service is that it provides real-time meteorological data that can be used in real-time air quality predictions.

The air quality data used in this investigation was acquired using rigorous quality assurance and control protocols, which include the followings.

- *Equipment Calibration:* Every air quality monitoring device used in this investigation underwent routine calibration in accordance with the instructions of the manufacturer. Prior to beginning data collection, calibration was performed to ensure the accuracy and reliability of the measurements. Furthermore, every three months the stations undergo a QA/QC and calibration. For $PM_{2.5}$, we use very precise automated German GRIMM equipment for calibration and QA/QC monitoring of data.
- *Site Selection:* The choice of monitoring locations was deliberated over thoroughly to provide an evenly distributed sample throughout all regions of interest. Considerations such as the location of emission sources, changes in land use, and population density informed the selection of the locations in order to capture spatial variations in air quality.
- *Quality Control Checks:* All collected data went through many rounds of rigorous quality control tests. The field personnel have been instructed on the normal procedures for operating and maintaining the equipment. The sensors were calibrated, the zero drift was checked, and the data was validated against established reference values to ensure the monitoring equipment were operating properly.
- *Data Validation and Quality Assurance:* Data validation and quality assurance procedures were implemented to make sure the information was accurate and trustworthy. This involved the identification and removal of any outliers or anomalous data points. Data completeness and consistency were carefully examined to minimize errors and ensure the integrity of the dataset.
- *Data Management:* The collected data were stored in a secure and organized manner, with the necessary backups in place to prevent data loss. To maintain data integrity, data management protocols were followed, including timestamping, and metadata documentation.

2.3. Methods

2.3.1. Data pre-processing

2.3.1.1. Removing unrealistic values. Initial data analysis revealed that the sensors recorded exaggerated $PM_{2.5}$ values at times. For example, the sensors recorded extremely higher concentrations of $PM_{2.5}$ (>180000 μg/m^3) or negative concentrations (<0 μg/m^3), both of which are not achievable in reality. Therefore, those values were removed and replaced with 'nan' indicating missing observations in the dataset.

2.3.1.2. Frequency conversion. Initially, $PM_{2.5}$ values were measured at 1-min intervals (every minute, sensors measure concentrations of $PM_{2.5}$ at the Healthyair air quality stations). The Darksky meteorological data, on the other hand, were recorded hourly. Therefore, down-sampling was used to transform the 1-min frequency of $PM_{2.5}$ concentrations into the hourly time scale by aggregating the mean of a specific hour (i.e., 00:00:00 to 00:59:00) and then merging it with meteorological data.

2.3.1.3. Removing outliers. All excessively high $PM_{2.5}$ levels are most often the result of irregular events (such as the large amounts of solid waste burning in the open air, the big number of fireworks used in events, few hours of road traffic jamming due to faulty vehicles). In this study, we have defined the outlier's threshold i.e., hourly $PM_{2.5}$ values that are both higher than 200 μg/m^3 and are three times higher than the $PM_{2.5}$ values of the previous hours are considered outliers. Our goal is to detect unrealistic values, so we implemented two conditions that must both be satisfied simultaneously. For instance, if the current $PM_{2.5}$ value at 12:00 p.m. is 300 (μg/m^3), which is higher than 200 and three times greater than the previous value of 100 (μg/m^3) at 11:00 a.m., then both conditions are satisfied, and it will be considered an outlier. Because such a rapid increase (e.g., from 100 to 300 within an hour) can be unrealistic in an urban setting (except for extreme events, which we have not considered in our study).

In another case for example, if the current $PM_{2.5}$ value at 9:00 a.m. is 400 (μg/m^3), which is higher than 200 (μg/m^3) but not three times greater than the previous value of 350 (μg/m^3) at 8:00 a.m., then only one condition is satisfied, and it will not be considered an outlier. This smaller increase can be attributed to peak or rush hours.

Finally, all outliers were removed and replaced with 'nan' which were considered missing values.

2.3.1.4. Missing data handling. Missing data occurred at all stations, primarily because of temporary power outages or critical malfunctions. We replaced missing data for a specific hour with the previous hour's concentration, and we replaced missing values detected in multiple consecutive hours with the concentrations of the same hours the previous day. Only in the training datasets were missing values supplied, but not in the testing datasets. As a result, the performance of the model given in this study is tested only using real-world testing data.

2.3.2. Correlation analysis

We applied both linear and non-linear techniques to identify the correlations between air pollutant $PM_{2.5}$ and meteorological variables. This is because linear correlation approaches may not always be adequate to effectively determine the association between two variables. This occurs specifically when the target variable ($PM_{2.5}$) is non-stationary and the level of uncertainty is high.

The Pearson correlation coefficient approach was utilized to examine the linear relationship between the hourly concentration of $PM_{2.5}$ and meteorological data in this study.

The linear statistical approach can only be effective to detect linear correlations among the variables, but it can be false in circumstances when the variables are associated with each other in a non-linear fashion. Therefore, to discover and evaluate the non-linearity among our variables, we also employed Mutual Information (MI) correlation analysis (Laarne et al., 2021). Section 3.2 provides a thorough explanation of correlation analysis.

2.3.3. Model development

2.3.3.1. Feature generation. Creating additional new features, including rolling mean and temporal features based on forecasting goals, is an important step toward enhancing model accuracy. Rolling-mean features were constructed utilizing real previous observations of raw $PM_{2.5}$ at precise intervals such as $PM_{2.5}$_3 h, $PM_{2.5}$_6 h, $PM_{2.5}$_9 h, $PM_{2.5}$_12 h, $PM_{2.5}$_15 h, $PM_{2.5}$_18 h, $PM_{2.5}$_21 h, and $PM_{2.5}$_24 h.

The primary aim for creating rolling-mean $PM_{2.5}$ features is to transform raw $PM_{2.5}$ observations into a more linear form, reducing the amount of uncertainty in $PM_{2.5}$ data at different intervals and providing long-term data patterns to the linear model.

Our analysis showed that $PM_{2.5}$ is highly seasonally dependent. Therefore, temporal variables, specifically 'month' and 'hour' can also be helpful for model development. However, we have data from February 2021 to September 2022, which includes the Covid-19 lockdown in 2021. As a result, we cannot use a monthly feature since the air quality of $PM_{2.5}$ during the regular and lockdown periods would change. Therefore, we only included the temporal parameter 'hour' when developing the $PM_{2.5}$ forecasting model.

2.3.3.2. Model construction and evaluation. A multivariate multi-step $PM_{2.5}$ forecasting model was built using $PM_{2.5}$ data, meteorological circumstances, and temporal factors. The input and output data were generated in a supervised learning fashion to implement the supervised machine learning algorithm.

Input pair (X) consistent of past 48-h ($t-47, t-46, t-45, \ldots, t-1$, t) concentrations of the following variables:

- $PM_{2.5}$ and rolling mean variables generated from raw $PM_{2.5}$ including $PM_{2.5}$_3 h, $PM_{2.5}$_6 h, $PM_{2.5}$_9 h, $PM_{2.5}$_12 h, $PM_{2.5}$_15 h, $PM_{2.5}$_18 h, $PM_{2.5}$_21 h, and $PM_{2.5}$_24 h.
- Meteorological variables (temperature, humidity, dew point, and wind speed) selected based on our correlation analysis
- Temporal variable 'hour'.

Output pair (y) consistent of future 24-h ($t+1, t+2, t+3, \ldots, t+23, t+24$) concentrations of $PM_{2.5}$ were predicted by the model. In this paper, the daily forecasting was not useful because of the large variability of $PM_{2.5}$ observed during the day (from 0 to 23 h). Another reason was that the daily averaged $PM_{2.5}$ forecasting model was not effective in reality as it gave only averaged concentration across the day and people cannot understand at which hours the level of air pollution will be high at a particular location in the city. Therefore, we proposed an hourly $PM_{2.5}$ forecasting model to help citizens plan efforts to alleviate the harmful effects of air pollution on health. The forecasting period was set to 24 h since long-term predictions such as 48 h, 72 h, and 96 h were not feasible owing to insufficient historical data available for a year which included the Covid-19 pandemic. Furthermore, according to the established training protocol, short periods of data (approximately three months between iterations) were employed in the training phase to address the non-stationarity issue of the $PM_{2.5}$ time series. Consequently, the results produced for predicting lengthy periods utilizing short periods of historical data were unreliable.

The optimal length of input (48 h) was determined by our experiments carried out on different lengths of input ranging from 12 h to 96 h. Appendix A (Table 2) has a description of the features utilized during model construction.

2.3.3.2.1. Data normalization. The data normalization was carried out after preparing the data in the appropriate input-output format. In model development, data normalization assigns equal weight to each contributing variable. The training dataset was scaled using a min-max scalar (Eq. (1)) to transform variables to ranges between 0 and 1 throughout the entire dataset.

Having all of the variables on the same scale helps the machine learning model to improve its forecasting accuracy. During the model performance evaluation, the reference of scalar objects utilized in the training set was also employed in converting the testing dataset on the same scale.

$$x' = \frac{x - min(x)}{max(x) - min(x)} \quad \text{Eq. 1}$$

2.3.3.2.2. Modelling method. In this study, we used LightGBM (Shi, 2007), a gradient-boosting framework based on decision trees, to develop a multivariate time series $PM_{2.5}$ forecasting models. We chose LightGBM because it often outperforms LSTM (Long Short-Term Memory) networks in scenarios where tabular data and interpretability are crucial (Shwartz-Ziv and Armon, 2022). Another reason for choosing the tree-based model is that LSTM requires more historical data to train effectively and capture temporal dependencies. However, during our experiment phase, the historical data was limited since we had just established our own network to collect the data. Additionally, in our previous article (Rakholia et al., 2022), we conducted a comparative experiment between a tree-based model (XGBoost) and an LSTM model. Our findings indicated that XGBoost slightly outperformed the LSTM.

LightGBM is highly efficient, offering faster training times and better handling of missing values compared to LSTM and it excels at capturing interactions between features without requiring extensive feature engineering, making it a robust choice for many real-world applications, such as predicting $PM_{2.5}$ levels from environmental data. Therefore, LightGBM is preferred in our situation, where speed, scalability, and interpretability are prioritized over the deep learning capabilities of LSTM.

In contrast to other tree-based algorithms, which grow trees by depth level, LightGBM develops tree leaf-wise in terms of best-first. This is done largely to optimize model training speed, accuracy, and memory utilization. (Shi, 2007). This will select the leaf with the greatest delta loss to expand the tree, resulting in a smaller loss than level-wise tree-based methods. Many tree-based boosting models (such as the default method in "XGBoost") employ presort-based techniques for learning in the decision tree, which is easier in some cases but difficult to optimize. In this case, LightGBM employs histogram-based algorithms to generate discrete bins from the values of continuous features, therefore increasing compute power (in terms of model training) and decreasing memory use.

When the dataset size is small, the leaf-wise tree structure might lead to model overfitting. To get over this limitation, we restrict the depth of the decision tree by setting the "max depth" parameter to a small number (the optimal small value based on testing). The tree may still expand in a leaf-wise direction even if the "max depth" parameter is supplied. Appendix A (Table 3) lists the parameter settings that were used to build the $PM_{2.5}$ forecasting model. The model parameters were tuned through trial-and-error experiments, using the correlation score and Mean Absolute Percentage Error (MAPE) as evaluation metrics.

2.3.3.2.3. Performance evaluation. This paper explores the problem of $PM_{2.5}$ prediction using a regression approach that is concerned with how predicted values are closer to the actual values of $PM_{2.5}$. To understand the average error across all experiments, the statistical evaluation measures Root Mean Squared Error (RMSE), MAPE, Mean Absolute Error (MAE), and correlation have been selected. Let $\hat{y}_i$ be a predicted value and y_i be an actual observation of $PM_{2.5}$, where $i = 1, 2, 3, \ldots, n$ and n be the number of samples in the testing datasets. The statistical formula of each evaluation indicator is given as follows:

$$RMSE\,(y, \hat{y}) = \frac{\sum_{i=1}^{n_{testing\_samples}} (y_i - \hat{y}_i)^2}{n_{testing\_samples}} \quad \text{Eq. 2}$$

$$MAE(y, \hat{y}_i) = \frac{1}{n_{testing\_samples}} \sum_{i=0}^{n_{testing\_samples}-1} |y_i - \hat{y}_i| \quad \text{Eq. 3}$$

To calculate the error between actual and predicted values in terms of percentage, MAPE was used:

$$MAPE(y, \hat{y}) = \frac{1}{n_{testing\_samples}} \sum_{i=1}^{n_{testing\_samples}} \frac{|y_i - \hat{y}_i|}{max(\epsilon, |y_i|)} \quad \text{Eq. 4}$$

Where ϵ is a small, fixed positive value to avoid undefined results when the actual value of (y_i) is zero.

3. Results and discussion

3.1. P.M.$_{2.5}$ trend analysis

The formation of the legend shown in (Fig. 1) is as follows:

$PM_{2.5}$_StationNo_StationCategory in which (I stand for Industrial, R for Residential, and T for Traffic). Station number 1 and station number 6 belong to mixed geographical areas.

The trend analysis was performed primarily to understand the temporal features, which can be used in forecasting model development to improve the accuracy of the model. Fig. 1 shows that $PM_{2.5}$ was highly dependent on spatio-temporality and seasonality. The amount of air pollution during the day was higher than at night, and the air quality of $PM_{2.5}$ changed over time. Whereas during the winter season, the concentration of $PM_{2.5}$ was found to be higher than in the summer and monsoon seasons. Because in winter, often there was an inversion in the thermal layer where the air is deposited and cannot be dispersed into high layers of the atmosphere. Pollution trapped in the air layer close to the ground causes the $PM_{2.5}$ concentration higher due to less dispersion within the smaller layer height.

The level of air pollution was found to be different across the day (from 0 to 23 h) at different stations, but the same pattern occurred repeatedly over a period during the weekdays and weekends, which suggested that 'hour' of the day can be considered as a feature in forecasting model development. $PM_{2.5}$ was also highly dependent on monthly seasonality, but due to the lack of enough cyclic data for a few years, we cannot use the 'month' feature (as we only had the data for 16 months).

3.2. Correlation analysis

Fig. 2 shows the correlation matrix for our linear correlation analysis. $PM_{2.5}$ had a negative association with dewpoints and wind speed below the moderate level (−0.5), but a mild positive link with the UV index. The Pearson correlation approach revealed no associations between $PM_{2.5}$ and other meteorological data such as temperature, humidity, and visibility. But past evidence showed temperature and humidity have a significant impact on affecting air quality $PM_{2.5}$ (Wang and Ogawa, 2015; Yang et al., 2017; Anusasananan et al., 2021).

The results of our MI correlation analysis are shown in Fig. 3. MI correlation generates a non-negative correlation strength range between [0 (low) to 1 (high)] instead of finding a positive or negative correlation. The scale of the MI correlation coefficient is non-linear and consequently extremely low correlation values may be strongly affected by random noise (Laarne et al., 2021). Due to the random noise in extremely low correlation coefficient values, a negative correlation was detected at several spots in Fig. 3. According to unconditional MI (Fig. 3-a), there was a weaker correlation between $PM_{2.5}$ and the meteorological variables temperature, humidity, dewpoint, wind speed, ultraviolet index, and visibility, as well as the temporal variable hour, which was not found in the linear correlation matrix for temperature, humidity, and visibility variables (Fig. 2)

MI correlation can also be utilized to explain the relationship between two variables based on their conditional dependence on another variable. In Fig. 3 (a), for example, unconditional MI-based correlation was estimated utilizing all variables. In contrast, conditional dependence was evaluated for the variable 'hour' in Fig. 3 (b). When the effectiveness of the variable 'hour' is reduced (Fig. 3 - b), the correlation between $PM_{2.5}$ and hour is about zero (not exactly zero due to quite significant noise in the data), and the correlation between $PM_{2.5}$ and temperature, humidity, dewpoint, and wind speed was found to be stronger than the correlation found in unconditional MI analysis (Fig. 3 - a).

The results suggest that the variables temperature, humidity, dewpoint, and wind speed explain variation within the hours during the day. The correlation between $PM_{2.5}$ and visibility and the UV index, on the other hand, was slightly smaller than that obtained in (Fig. 3 - a), showing that the hours' cycles of those variables were partially responsible.

3.2.1. Time dependency MI correlation of $PM_{2.5}$

To construct a time series forecasting model for $PM_{2.5}$, it is necessary to understand how $PM_{2.5}$ is affected by prior measurements of weather conditions. Time dependent MI correlations help to estimate the

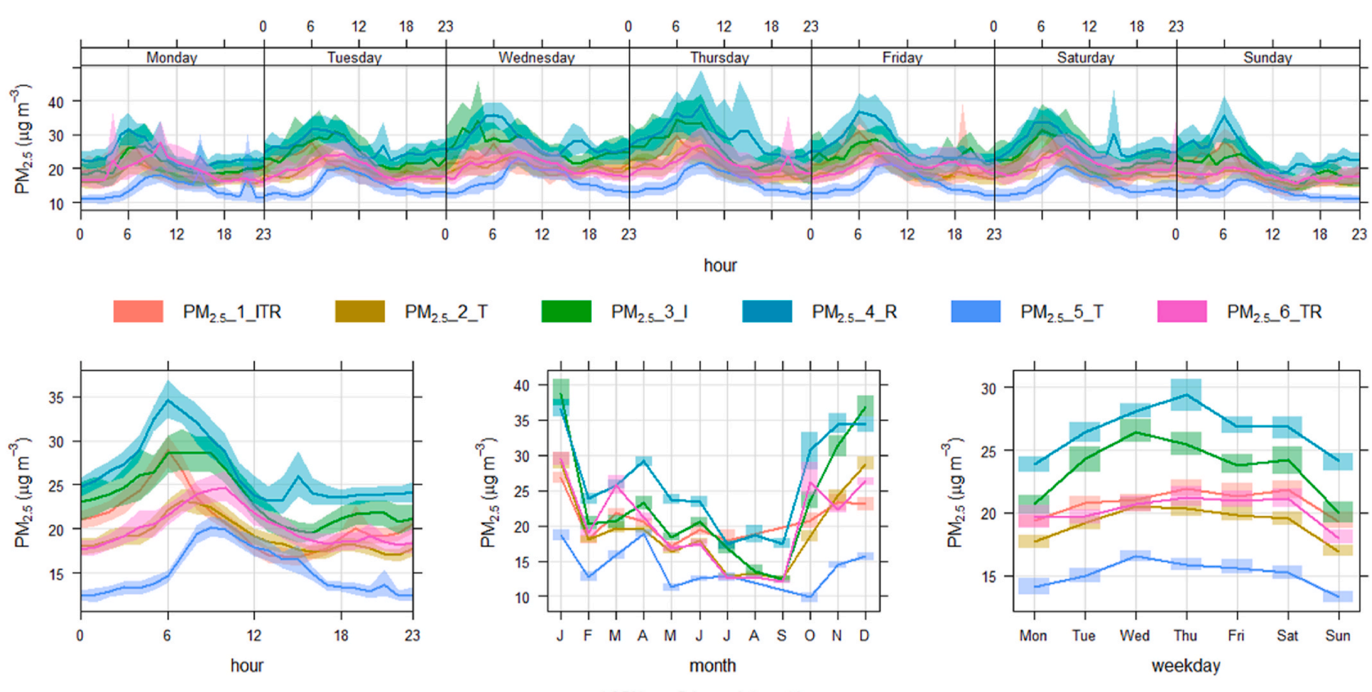

Fig. 1. Trend analysis of $PM_{2.5}$ time series in μg/m^3. The left bar has an average range of $PM_{2.5}$ from March 2021 to September 2022. The legend ITR denotes Industrial, Traffic and Residential.

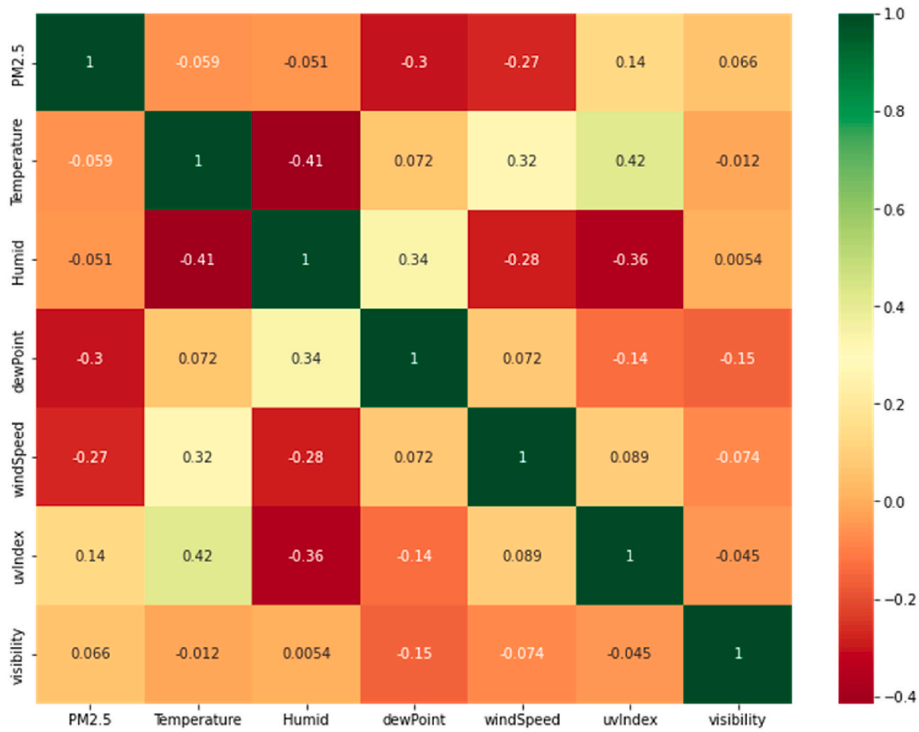

Fig. 2. Statistical Pearson correlation matrix (PM2.5 + Meteorological variables.

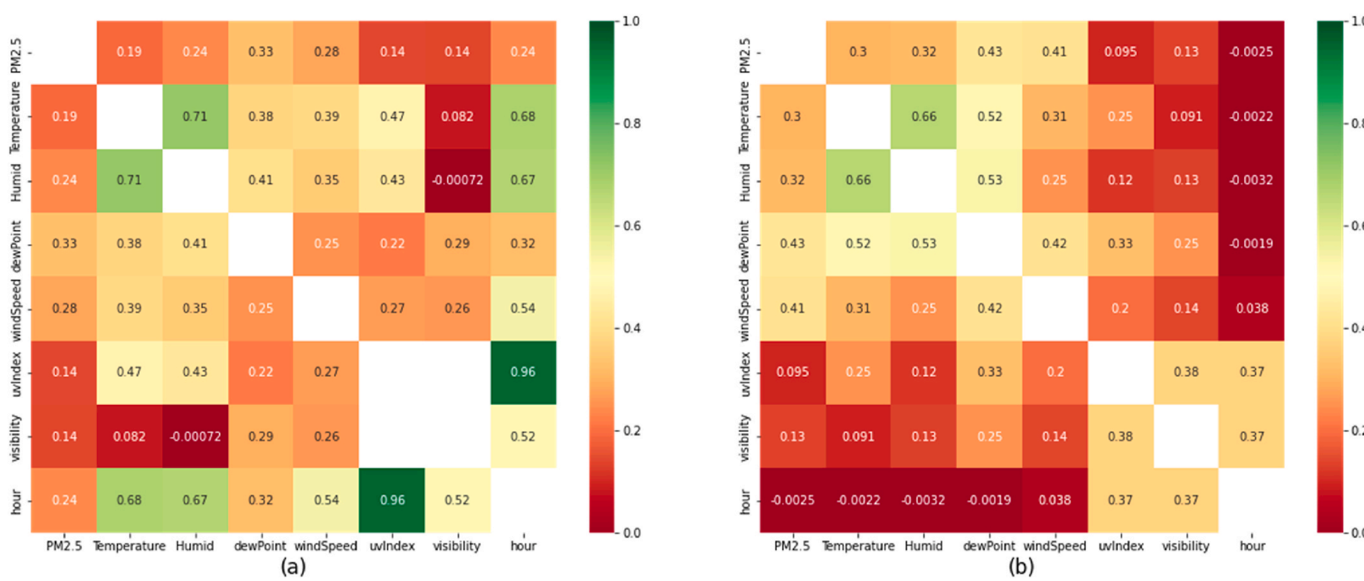

Fig. 3. MI-based correlation matrices (a) including temporal variable (b) removing effectiveness of temporal variable.

correlation between the concentration of $PM_{2.5}$ now and the past [0 h–48 h in our case] measured values of other variables.

Fig. 4 shows that, except for wind speed, the association between $PM_{2.5}$ and meteorological data was nearly stable, indicating that weather did not change drastically in the previous 48 h. The association between $PM_{2.5}$ and the dew point was stronger than the correlation between $PM_{2.5}$ and other meteorological indicators, but visibility and UV index were shown to be moderately associated with $PM_{2.5}$ lags during the previous 48 h. As a result, we did not include the climatic parameters visibility and UV index as lagged covariates when developing the model.

3.3. P.M.$_{2.5}$ data analysis and forecasting results

In this study, we focused on exploring the opportunities for effectively increasing the accuracy of the multi-step hourly $PM_{2.5}$ forecasting model. Based on our correlation analysis, the meteorological variables used as inputs for our forecasting models include Temperature, Humidity, Dewpoint, and Wind speed. The hourly $PM_{2.5}$ concentrations and meteorological conditions were used to develop a multi-step multivariate forecasting model to predict the concentrations of $PM_{2.5}$ in the next 24 h. Exploratory data analysis Appendix B (Fig. 3) and trend analysis (Fig. 1) indicated that the degree of air pollution varied among the six sites, implying that training a single model utilizing aggregated $PM_{2.5}$ data from all stations would be insufficient. We trained and

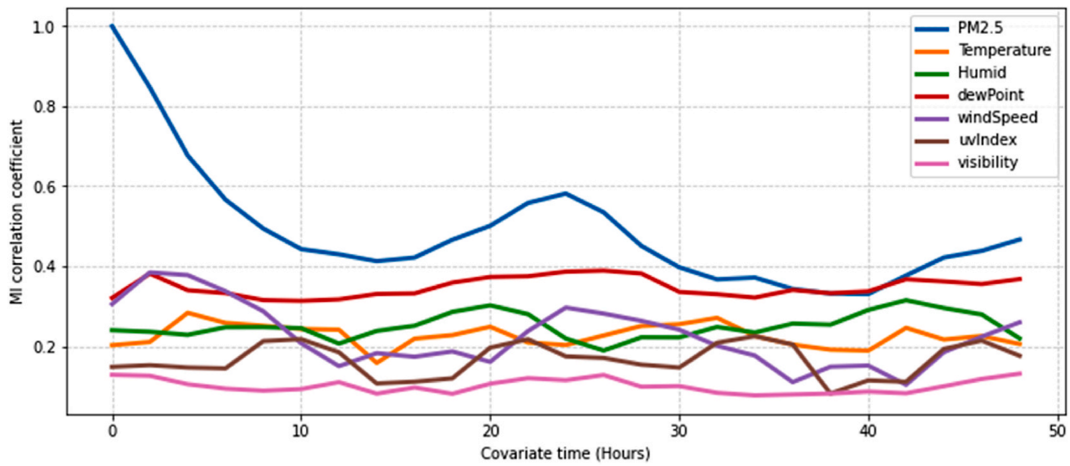

Fig. 4. Time dependency MI correlation of $PM_{2.5}$ with meteorological variables based on the past 48 h' concentrations.

evaluated six forecasting models, one for each site in this investigation. In this study, the characteristics of the data for each site differ because our monitoring sites are categorized into different areas such as traffic, residential, and industrial zones. Therefore, constructing a single model for all sites is not a good choice for ensuring consistent performance over time. Additionally, the data is non-stationary, with its mean and variance changing at different points in time, making it difficult for one model to adapt to these changes.

The brief overview of $PM_{2.5}$ data included data distribution, percentage of missing data, percentage of available data in each month across all stations, and the value of some statistical measures are displayed and described in Appendix B (Fig. 3).

The maximum percentage of missing data was recorded by stations number 1 and 5 compared to other stations during the lockdown periods of Covid-19. During that period station 1 and station 5 were shut down due to having frailer power as well as some technical issues, and during that period no technical supports were available in the city due to the total lockdown.

Based on the availability of air pollution data for a short period of time (February 2021 to September 2022), defining a suitable model training protocol is a crucial part of the $PM_{2.5}$ forecasting model development. Especially since the data used in this research was collected during the exceptional months of the Covid-19 lockdown. Therefore, the model trained on the mixed periods of data and forecasting for a normal period will not be reliable. Here, we have designed a new model training protocol considering the non-stationarity of the $PM_{2.5}$ time series and kept testing examples closer to the training data. We trained each forecasting model using the data for a period of three months and the partition of training and testing used in model construction was 90:10 (90% for training and 10% for testing). Furthermore, the experiments were carried out for model training using different slices of the dataset created by shifting a sliding window of one week through the entire dataset. Whereas, 10% of testing data were used for model evaluation and performance measurement in which each forecast was generated by moving a sliding window of 1 day (24 h) through the entire testing set. Using the entire dataset, the model was trained 56 times using the proposed training protocol and each time 10% of the data was used for model evaluation.

Additionally, we have also defined a threshold of 70% to verify that a reasonable number of data were available for model training in each chosen slice of data for each station. If the amount of data available for training was less than the defined threshold of 70%, then that slice of data was skipped and not considered for model training. Finally, the results were aggregated by mean for all experiments.

The results of the multi-step $PM_{2.5}$ forecasting model (considering meteorological conditions, rolling mean, and temporal features) are presented in Table 1. The results show that the lowest RMSE of 4.85 μg/m^3 and MAE of 3.36 μg/m^3 were reported for station 5 and station 6 obtained similar results in terms of MAPE compared to station 5. The highest correlation was reported at 0.77 by station 6 followed by station 5 (0.72), and then stations 2 and 4 (0.71). The performances varied station by station, with the highest value of RMSE reported by stations 3 and 4. The reason might be the higher concentrations of $PM_{2.5}$ observed there compared to other stations (Fig. 1).

The averaged performance of the $PM_{2.5}$ forecasting model across all stations with associated statistical measures is presented in Appendix A (Table 4). These results prove that the model developed in this study is performing well for air pollution prediction, i.e., the model has a correlation value greater than 0.5. The time series comparison between the predicted and observed $PM_{2.5}$ levels for each station is shown in Fig. 5.

3.4. *Analyzing the contributions of different feature groups with respect to forecasting model performance*

To understand the contributions of rolling mean and temporal features incorporated in forecasting model development we have also trained the model without considering rolling mean and temporal features. The model was trained, tested, and evaluated using the same protocol for comparison purposes. The results generated without using rolling mean features are presented in Table 2 with associated aggregated statistical measures are presented in Appendix A (Table 5). Compared to Table 1, the results clearly shows that the rolling mean and temporal features make a significant contribution with respect to increasing the accuracy of the model.

Fig. 6 shows the correlation results across the forecasting horizon (1 h–24 h) for three types of forecasting models: multivariate model with rolling mean features, models without rolling mean features and univariate models. The results clearly reveal that rolling mean features significantly contribute to forecasting model accuracy; models with

Table 1
Averaged 24-h $PM_{2.5}$ forecasting results considering rolling mean features of $PM_{2.5}$.

Station Number	RMSE (μg/m^3)	MAPE (0-low, 1-high)	MAE (μg/m^3)	Correlation (0-low, 1-high)
1	7.22	0.27	5.32	0.70
2	7.72	0.23	4.76	0.71
3	12.34	0.33	8.01	0.69
4	9.83	0.24	6.37	0.71
5	4.85	0.20	3.36	0.72
6	5.94	0.20	3.94	0.77

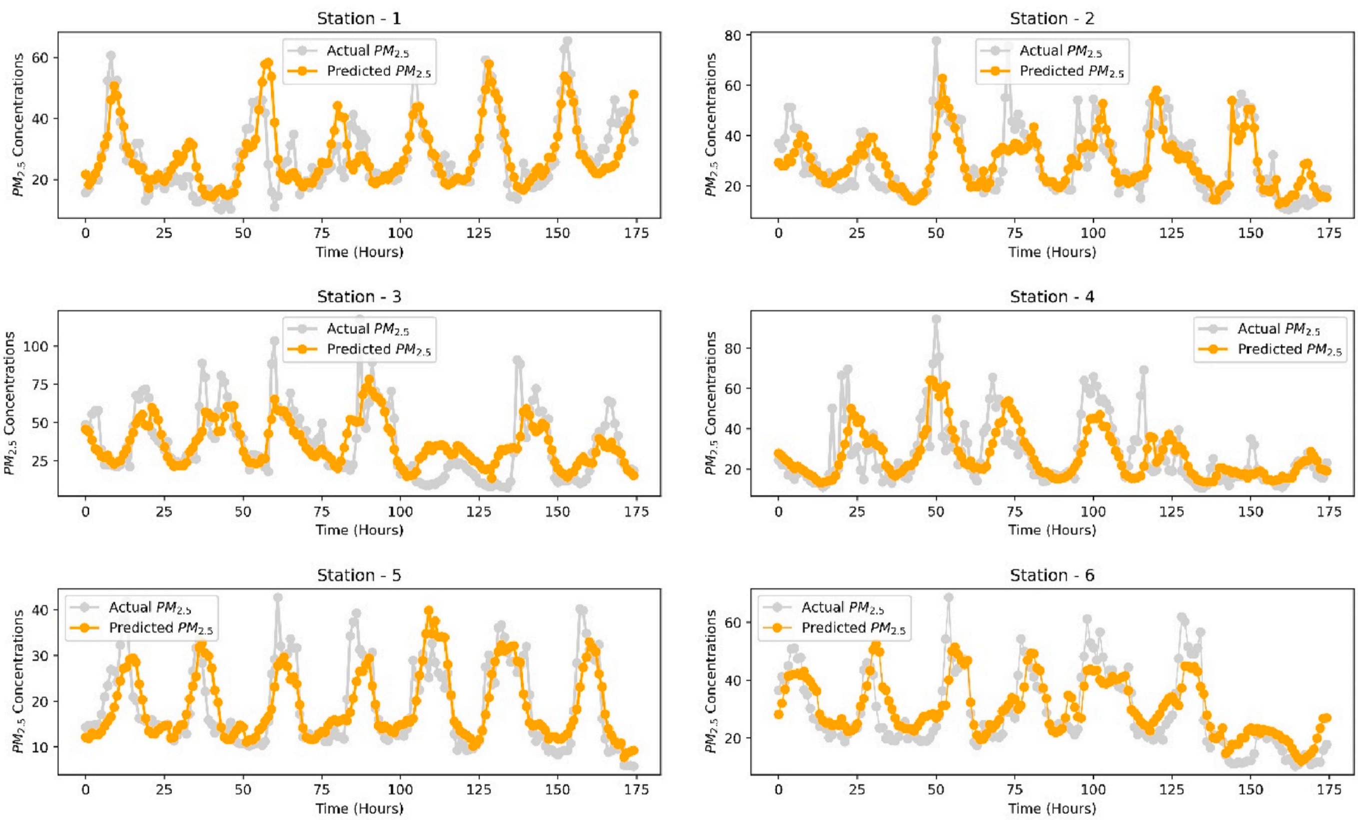

Fig. 5. Time series comparison between the predicted and observed PM2.5 levels.

Table 2
Statistical analysis on averaged $PM_{2.5}$ forecasting results across six stations (without rolling mean features).

Station Number	RMSE (μg/m^3)	MAPE (0-low, 1-high)	MAE (μg/m^3)	Correlation (0-low, 1-high)
1	9.58	0.54	7.12	0.34
2	9.47	0.34	6.35	0.49
3	16.63	0.54	11.09	0.50
4	15.67	0.39	9.44	0.37
5	7.30	0.31	4.89	0.46
6	10.03	0.31	6.0	0.44

rolling mean features performed better over the entire forecasting horizon.

3.5. Comparative analysis

In this experiment, we compared our proposed method against other frequently used methods for air quality forecasting. These included a XGBoost model, a two-layer LSTM model, and a hybrid model combining two layers of CNN (Convolutional Neural Network) and one layer of LSTM (CNN-LSTM). The aggregated results from all stations (Table 3) show that tree-based algorithms such as LightGBM and XGBoost consistently outperformed neural network-based algorithms like LSTM and CNN-LSTM, confirming our hypothesis that the limitation of our historical data restricts the performance of neural network models.

3.6. Significance of our work

- *Multivariate Ensemble Learning Approach*: We utilize a multivariate ensemble learning approach, specifically employing a tree-based ensemble machine learning method called LightGBM. Unlike traditional methods that grow trees by depth level, LightGBM develops trees leaf-wise using a best-first strategy. This approach significantly optimizes model training speed, accuracy, and memory utilization.
- *Accounting for Evolving Target Distributions*: Our model is designed to adapt to the evolving distributions of $PM_{2.5}$ concentrations. Unlike static models, our approach continuously updates and adjusts to changes in air pollution patterns, making it highly relevant for dynamic urban environments where pollution levels can fluctuate due to various factors.
- *High Prediction Accuracy*: Through extensive experiments and validations, our model demonstrates superior accuracy in forecasting $PM_{2.5}$ levels compared to existing methods. This high level of precision is crucial for timely and effective air quality management and public health interventions.
- *Comprehensive Data Utilization*: We incorporate a wide range of variables, including meteorological data, traffic patterns, and industrial activities, to capture the multifaceted nature of air pollution. This multivariate analysis provides a more comprehensive understanding and prediction of $PM_{2.5}$ levels.
- *Real-world Applicability*: Our model is designed with practical implementation in mind, providing actionable insights for urban planners, policymakers, and environmental agencies. The ability to accurately forecast $PM_{2.5}$ levels can significantly aid in decision-making processes and mitigation strategies.

4. Conclusion

This study aimed to assess the variation in $PM_{2.5}$ levels and understand the overall air quality across various regions in HCMC. To achieve this, we constructed a multivariate, multi-step forecasting model for $PM_{2.5}$ air quality. This model aims to empower citizens of HCMC by providing timely insights, enabling them to take immediate precautions

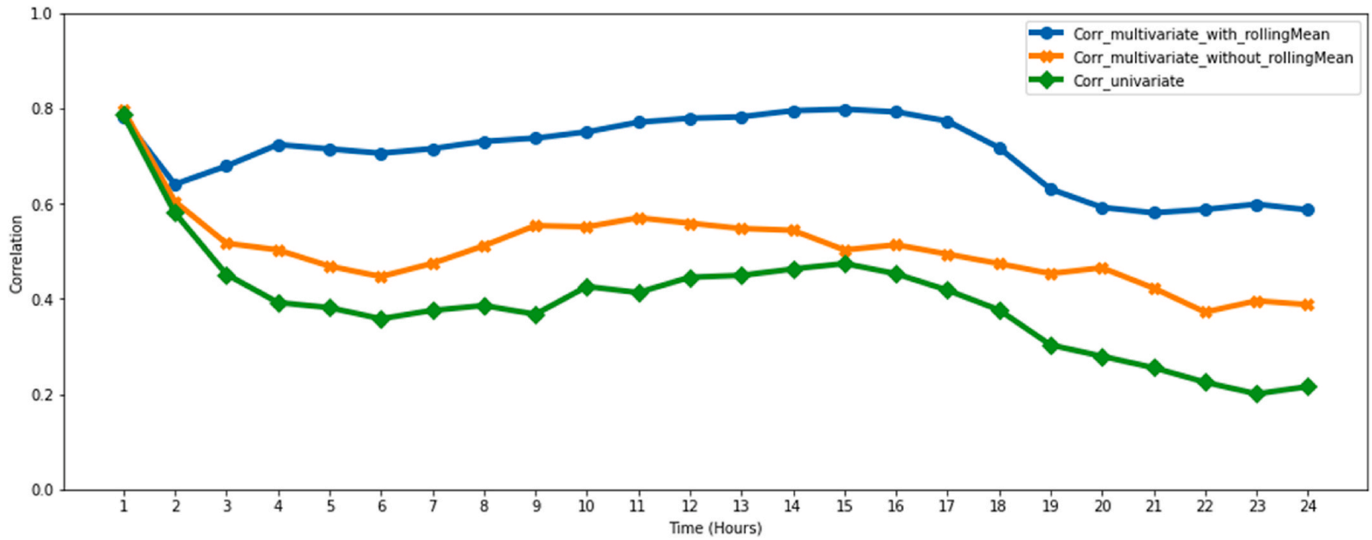

Fig. 6. Correlations among the models across forecasting horizon (1 h–24 h).

Table 3
Performance of other popular air-quality forecasting models. The results were aggregated across six monitoring stations.

Evaluation Matrix	LightGBM	XGBoost	LSTM	CNN-LSTM
Correlation	0.72	0.70	0.63	0.67
RMSE	7.98	8.20	11.03	9.98
MAPE	0.24	0.25	0.34	0.31
MAE	5.29	5.34	7.89	7.19

against harmful air pollution. Our experimental results validate the robust performance of our forecasting model, showcasing significant enhancements in forecasting accuracy compared to existing $PM_{2.5}$ models tailored for HCMC (Rakholia et al., 2022). Moreover, our model's versatility extends beyond HCMC; it can be adapted, trained, and applied in various cities across Vietnam and other developing nations. We employed a combination of linear and non-linear techniques to unveil the correlation between $PM_{2.5}$ levels and meteorological factors. Through meticulous feature engineering, including the integration of rolling features, we bolstered the model's predictive capabilities. Additionally, we conducted a thorough analysis of temporal variations to gain insights into the temporal dynamics of $PM_{2.5}$ and to create novel temporal features.

Our contribution.

- We integrated raw $PM_{2.5}$ rolling mean characteristics into our model, mitigating the impact of uncertainty and transforming raw $PM_{2.5}$ concentrations into a more linear form. This inclusion significantly enhanced the performance of the forecasting model.
- We employed both linear and mutual information-based nonlinear time-dependent correlation techniques to explore the relationship between $PM_{2.5}$ levels and meteorological factors.
- We implemented a training protocol for $PM_{2.5}$ forecasting models to account for the non-stationarity and seasonality across various pandemic phases, including periods before lockdown, during partial lockdowns, full lockdowns, and post-lockdown phases.
- To enhance the model's accuracy, we integrated temporal features by amalgamating historical PM2.5 data from diverse locations (such as industrial, residential, and transportation areas) with meteorological data.
- We incorporated a tree-based ensemble machine learning model into our framework, utilizing a leaf-wise construction approach instead of the traditional level-wise method. This enhancement aimed to improve model efficiency compared to previous tree-based models.
- We implemented a real-time $PM_{2.5}$ forecasting model within the Healthyair mobile application, providing valuable insights to citizens.

Limitations in the Study.

- Our model makes predictions for up to 24 h. It cannot make predictions for longer than 24 h because the training protocol was set up to keep the testing data close to the training data and only a limited amount of historical data was available for training in each iteration.
- Due to a lack of available wind direction data, this research did not examine or take into account air pollution that comes from different sources (locations).
- The model is only able to make a forecast if the historical input data are available for the last 48 h. This includes the $PM_{2.5}$ and weather parameters that were used to build the model. Otherwise, the model will return "no data found".
- While our air quality forecasting models achieved high accuracy across six locations with a fine temporal resolution, a few more steps will be needed to develop air quality forecasting models with a fine spatiotemporal resolution. These steps could be:1) increase the number of air quality stations installed in a geographical area, 2) modify the output of a forecasting model to be a multidimensional array (corresponding to different spatiotemporal dimensions), and 3) have ground-truth air quality data across space and time dimensions. We aim to tackle these issues with a new experiment design in our future research.

Broader impacts.

- This research demonstrated a variety of correlation methods applicable to analyzing relationships between variables in time series forecasting.
- This study's model training approach may be applied in similar situations when there is a scarcity of historical data on which to base a prediction model.
- By adjusting the model's parameters and retraining it, the same model may be used to predict $PM_{2.5}$ levels in different regions.

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- The Healthyair Project

This paper is an output from the Healthyair project which was funded by the Irish Research Council and the Department of Foreign Affairs in Ireland through the COALESCE call (Better World Awards, 2020) for reducing humanitarian needs, climate action, and strengthening governance. The project aims to assess the impact of air pollution and climate change on human health and prepare a policy for HCMC to reduce air pollution and improve population health. The project is a collaboration between CeADAR, Ireland's Centre for Applied Artificial Intelligence, based at University College Dublin (UCD), Ireland, and the Institute for Environment and Resources and the School of Economics and Law – both are based at Vietnam National University (VNU) in HCMC.

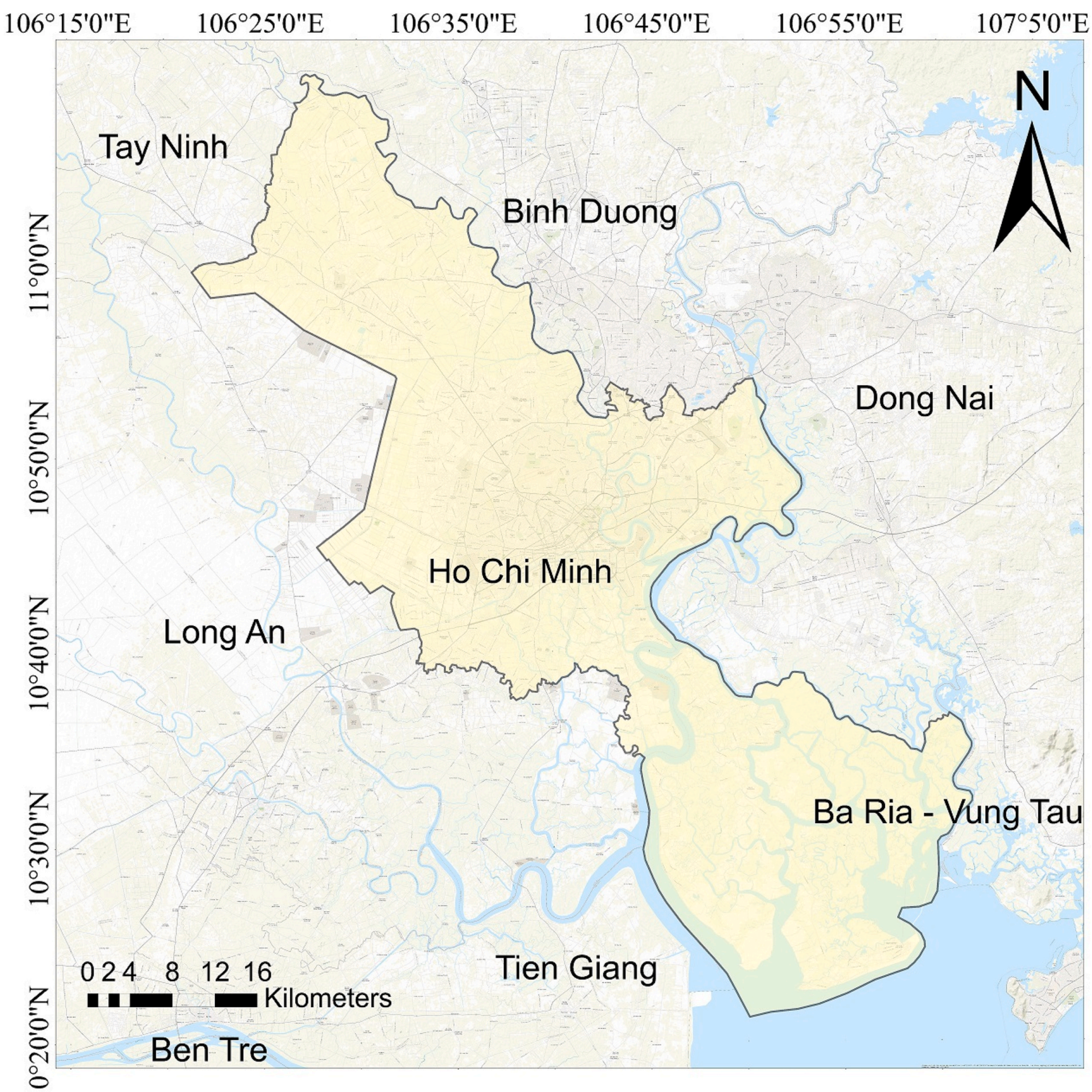

Fig. 1. Study area selected for the Healthyair research project

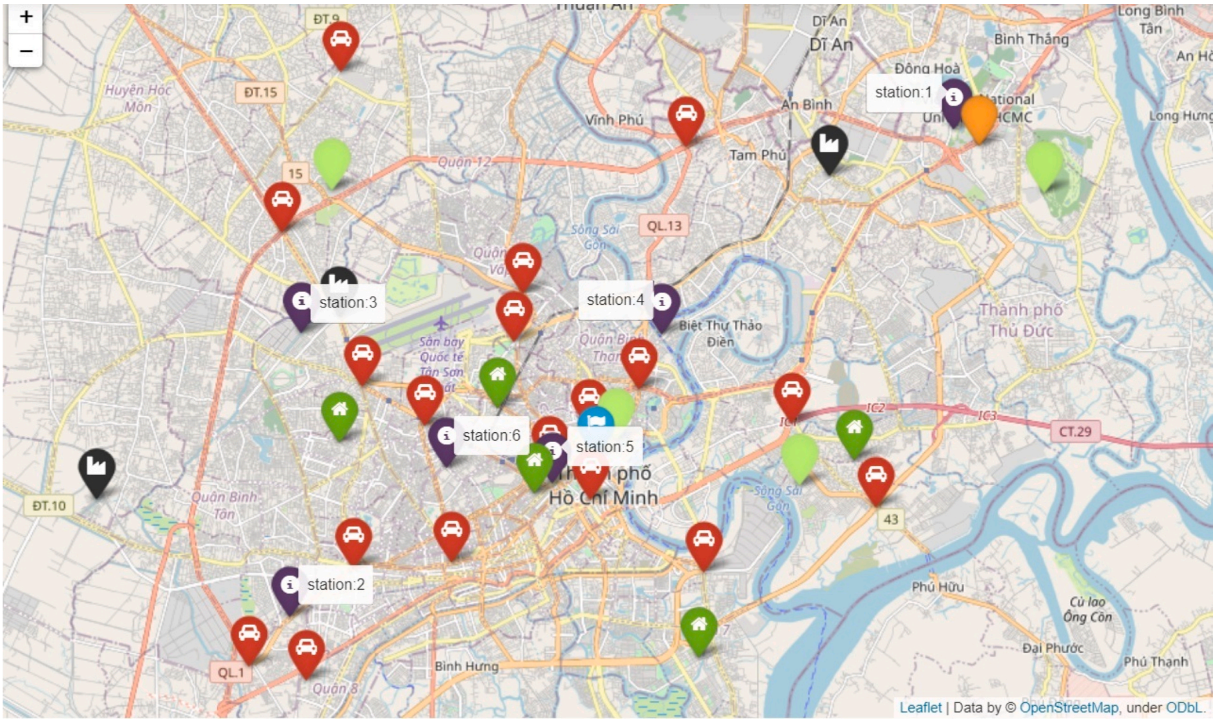

Fig. 2. Location of the six Healthyair Air Pollution Monitoring Stations installed in HCMC

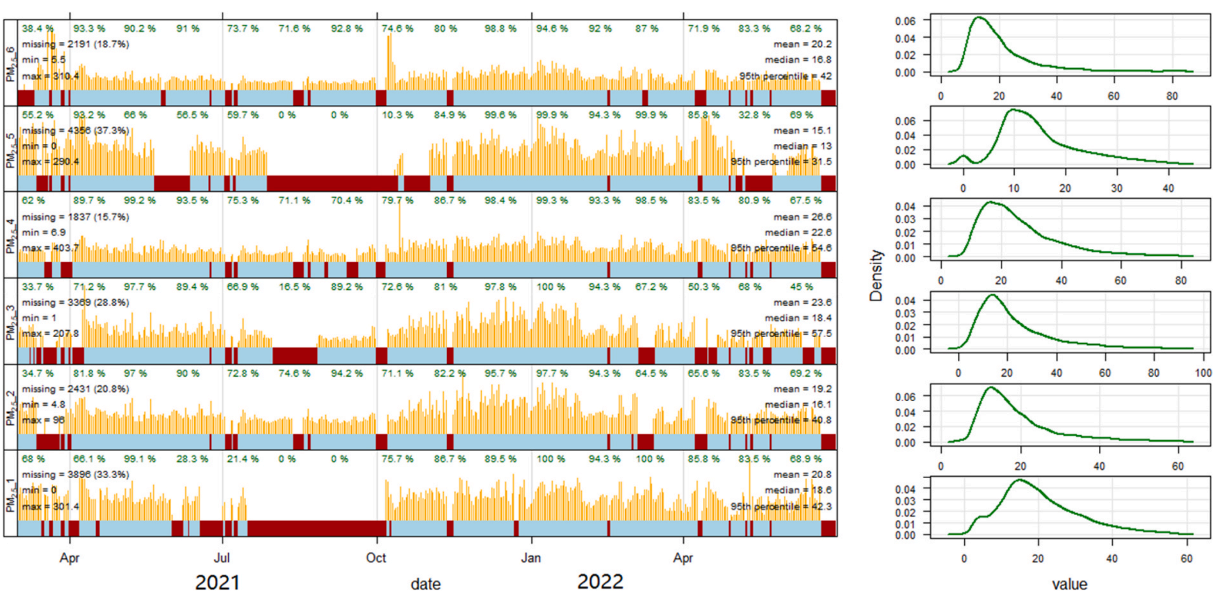

Fig. 3. Daily trends of PM2.5 across the adopted period of study

Table 1
Machine Learning Approaches used in PM2.5 Forecasting Model Development

References	Location	Study period	Data Sources	Sampling frequency	Prediction length	Methods and Results (RMSE) μg/m3
Rakholia et al. (2022)a	HCMC, Vietnam	February 2021 to December 2021	Healthyair air quality Monitoring stations in HCMC	1 Hour (mean)	1 h–24 h (One shot)	XGBoost (8.73) 1D CNN-LSTM (8.86) SGDRegressor (7.74) FB-Prophet (8.47)
Phung et al. (2020)	HCMC, Vietnam	2017	Generate inventory + data from	Not mentioned	Not mentioned	WRF + SMOKE + CMAQ

(continued on next page)

Table 1 (*continued*)

References	Location	Study period	Data Sources	Sampling frequency	Prediction length	Methods and Results (RMSE) μg/m3
Minh et al. (2021)	HCMC, Vietnam	2016 to 2021	US Embassy air quality station in HCMC + WRF's simulated meteorological data	24 h (mean)	1 day or up to 7days	Linear Regression: (7.80) Neural Network:(7.97 LASSO: (7.82) Extra Trees Regression:(7.68) Decision Forest Regression: (7.82) XGBoost: (8.30)
Guo et al. (2022)	Tianjin, China	July 1, 2020, to December 31, 2020	Generated through IoT devices	Not mentioned	Not mentioned	Bayesian optimization-based LSTM (13.06)
Perez et al. (2020)	Coyhaique, South America	2014 to 2016	Coyhaique II	1 h	24 h moving average	Neural network model and Linear model (RMSE not mentioned)
Zamani Joharestani et al., 2019	Tehran's urban area, Iran	January 2015 to December 2018	Retrieved from Tehran's Municipality ICT website	Daily mean	A day ahead	XGBoost (13.58)
Kumar et al., 2020	Delhi, India	January 2018 to November 2019	Air pollution monitoring station in 'R K Puram area', Delhi	Hourly data	Hourly basis (length not mentioned)	ET + AdaBoost (25.11)

Table 2
Variable descriptions

Variable name	Units	Description
$PM_{2.5}$	μg/m^3	Raw $PM_{2.5}$ values
Temperature	°C (Degrees Celsius)	Outdoor temperature
Humidity	Percentage (%)	Outdoor humidity: (Percentage of actual moisture content in gm^{-3})
Dewpoint	°C (Degrees Celsius)	Dew point
Wind speed	Meters per second	Wind speed
$PM_{2.5}$_3 h	μg/m^3	3-h rolling mean of $PM_{2.5}$
$PM_{2.5}$_6 h	μg/m^3	6-h rolling mean of $PM_{2.5}$
$PM_{2.5}$_9 h	μg/m^3	9-h rolling mean of $PM_{2.5}$
$PM_{2.5}$_12 h	μg/m^3	12-h rolling mean of $PM_{2.5}$
$PM_{2.5}$_15 h	μg/m^3	15-h rolling mean of $PM_{2.5}$
$PM_{2.5}$_18 h	μg/m^3	18-h rolling mean of $PM_{2.5}$
$PM_{2.5}$_21 h	μg/m^3	21-h rolling mean of $PM_{2.5}$
$PM_{2.5}$_24 h	μg/m^3	24-h rolling mean of $PM_{2.5}$
Hour	Integer number	An extracted hour from timestamps

Table 3
Parameter settings used in model development

Parameters	Values	Parameter Description
lags	48	Past 48 h concentrations (lags) of $PM_{2.5}$
lags_past_covariates	48	Past 48 h concentrations (lags) of covariate features (Temperature, Humidity, dewpoints, wind speed, and rolling mean features of $PM_{2.5}$)
n_estimators	120	Number of boosting trees used to fit the model
colsample_bytree	0.5	Percentage of the covariate features to be used for building each tree
min_child_weight	6	The process of further partitioning will be stopped if the sum of instances' weight is less than the specified value of min_child_weight
max_depth	6	To avoid deeper growing tree
learning_rate	0.15	That will determine the step size at each iteration while moving toward a minimum loss function.
objective	'poisson'	Name of the learning objective function to be used
metric	'rmse'	root square loss is considered as evaluation matrix
num_leaves	25	To control the complexity of the model. The values were set corresponding to the value of the 'max_depth' parameter.

Table 4
Statistical analysis on averaged PM2.5 forecasting results across six stations (with rolling mean features)

Statistics	Correlation (0-low, 1-high)	RMSE (μg/m^3)	MAE (μg/m^3)	MAPE (0-low, 1-high)
Mean	0.71	7.98	5.29	0.24
Std	0.10	3.39	2.19	0.07
Min	0.39	1.33	1.04	0.10
25%	0.65	5.89	4.09	0.20
50%	0.72	7.49	4.96	0.24
75%	0.78	9.84	6.47	0.27
Max	0.92	18.10	11.19	0.54

Table 5
Averaged 24-h PM2.5 forecasting results without rolling mean features of PM2.5

Statistics	Correlation (0-low, 1-high)	RMSE (μg/m^3)	MAE (μg/m^3)	MAPE (0-low, 1-high)
Mean	0.45	11.44	7.48	0.40
Std	0.14	4.58	2.72	0.12
Min	0.11	3.40	1.83	0.12
25%	0.36	8.12	5.41	0.32
50%	0.44	10.38	6.76	0.36
75%	0.54	13.50	9.24	0.48
Max	0.85	26.51	14.46	0.67

CRediT authorship contribution statement

Rajnish Rakholia: Writing – original draft, Methodology, Data curation. **Quan Le:** Writing – review & editing, Validation, Supervision, Conceptualization. **Khue Vu:** Visualization, Investigation. **Bang Quoc Ho:** Project administration, Funding acquisition, Conceptualization. **Ricardo Simon Carbajo:** Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

We have published our preprocessed dataset at https://data.mendeley.com/datasets/pk6tzrjks8/1

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